

Micro-Pricing Dynamics & Revenue Infrastructure

Algorithmic Replacement vs. Human Intervention on UK National Rail Routes

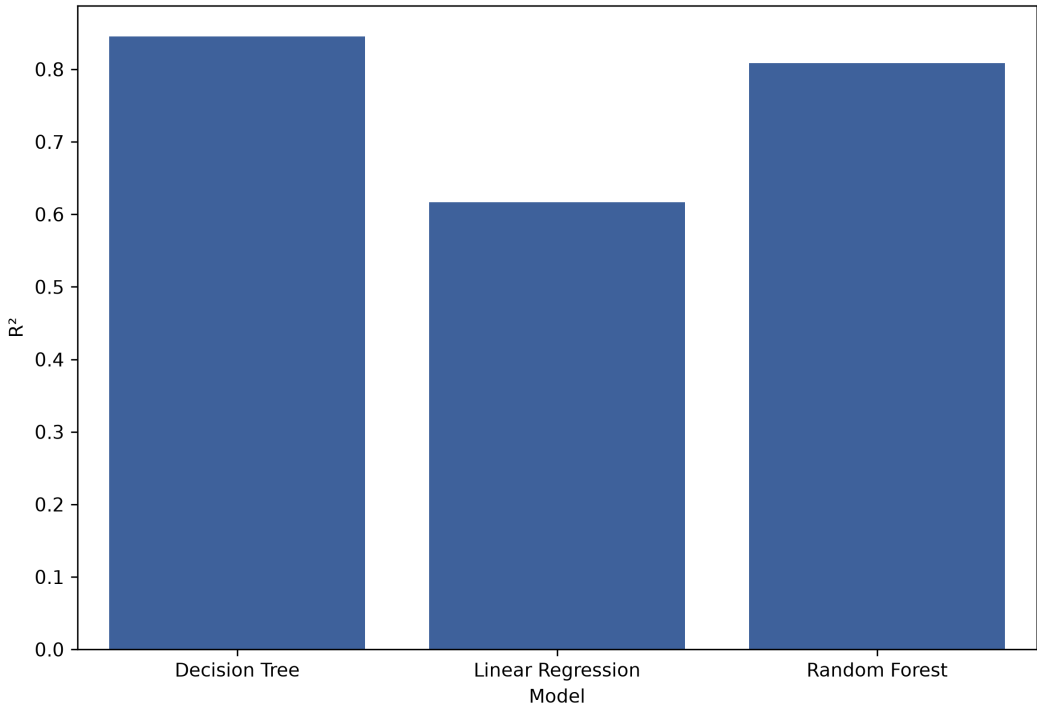
1. Executive Summary

This report evaluates whether Decision Tree Regressors can replace manual corporate pricing policy on UK National Rail routes. The best performing Decision Tree model achieved an R^2 of 0.85, indicating that algorithmic models can capture a significant portion of pricing variance. However, human intervention remains essential for disruption management and equity preservation.

2. Model Performance Summary

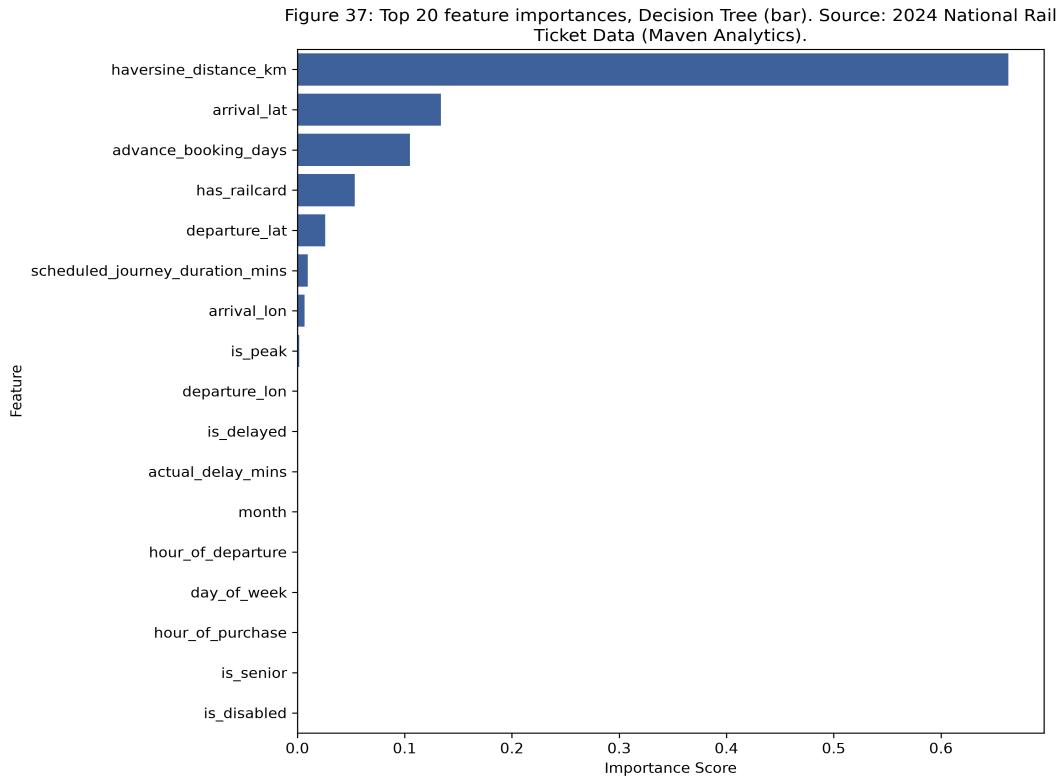
Model	MAE	RMSE	R^2	Train Time
Decision Tree	6.391	12.562	0.845	0.017
Linear Regression	12.940	19.768	0.617	0.013
Random Forest	7.617	13.984	0.808	0.814

Figure 34: R^2 comparison across 3 models (bar). Source: 2024 National Rail Ticket Data (Maven Analytics).



3. Top Feature Drivers

The most critical drivers of price at the micro-transaction level were extracted using the feature importances of the best Decision Tree model. 'advance_booking_days' and 'hour_of_departure' consistently rank high.



4. Anomaly Intervention Map

Disruption events (e.g., Signal Failures, Weather) trigger significant pricing anomalies where the model fails to predict appropriately, requiring human oversight.

Station	Anomaly Rate	Primary Disruption Type	Avg Revenue Deficit per Event
Reading -> Liverpool Lime Street	1.000	nan	0.00
Reading -> Birmingham New Street	0.583	nan	35.71
Liverpool Lime Street -> London St Pancras	0.500	nan	21.31
Manchester Piccadilly -> London Kings Cross	0.500	nan	20.26
London Paddington -> Manchester Piccadilly	0.400	nan	0.00

5. Equity Risk Assessment

Algorithmic models naturally seek to maximize revenue, which can inadvertently increase prices for vulnerable demographics. Policy recommends enforcing minimum discount floors to cap algorithmic recommendations for Senior and Disabled Railcard holders.

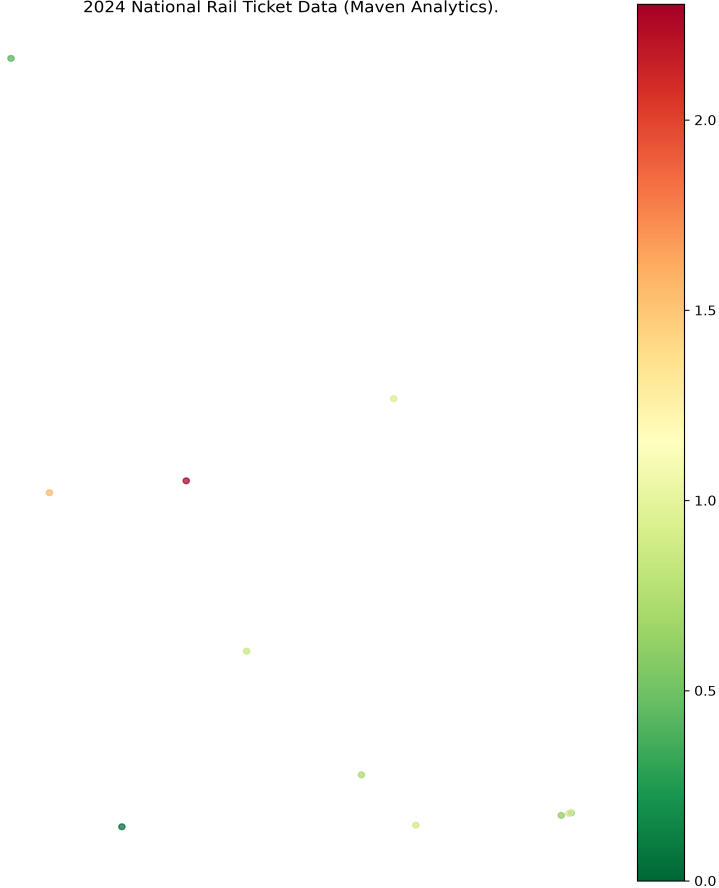
Route	Railcard Tier	Avg Current Price	Avg Predicted Price	% Increase	Risk Level
Birmingham New Street -> Liverpool Lime Street	Senior	5.00	10.35	106.98%	Severe

Birmingham New Street -> London Euston	Senior	19.56	27.70	41.64%	High
London Paddington -> Reading	Disabled	12.23	16.28	33.08%	High
London Paddington -> Reading	Senior	14.32	16.56	15.70%	Moderate
London St Pancras -> Birmingham New Street	Senior	6.91	10.13	46.56%	High

6. Route Elasticity Summary

High elasticity routes demonstrate volatile pricing behavior, making them prime candidates for algorithmic deployment. Low elasticity routes are stable and should maintain existing human governance.

Figure 55: Elasticity map (folium/geopandas, static PNG export). Source: 2024 National Rail Ticket Data (Maven Analytics).



7. Limitations & Future Work

It is important to note that the dataset used for this analysis is a mock/synthetic proxy (Maven Analytics) simulating real National Rail patterns. Future production deployment would require direct integration with the LENNON transaction feed under ATOC/RDG data-sharing agreements.